

Unit III: Magnetic Effects of Current and Magnetism

CHAPTER-5: MAGNETISM AND MATTER

GIST OF THE CHAPTER:

Bar magnet, bar magnet as an equivalent solenoid (qualitative treatment only), magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only), torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only), magnetic field lines.

Magnetic properties of materials- Para-, dia- and ferro - magnetic substances with examples, Magnetization of materials, effect of temperature on magnetic properties.

CONTENT/ CONCEPTS-

1. Bar magnet

It is a rectangular shaped small piece of a rod having south and north poles of same strength separated by a small distance.

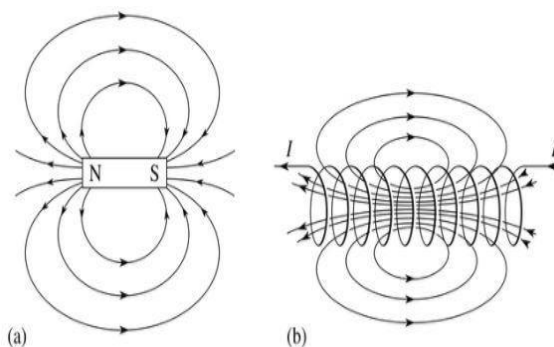
Properties :

- (i) Property of attraction.
- (ii) Property of alignment.
- (iii) Like magnetic poles repel each other, while unlike magnetic poles attract each other.
- (iv) Property of pairs
- (v) Property of testing
- (vi) Property of induction.

2. Bar magnet as an equivalent solenoid (qualitative treatment only)

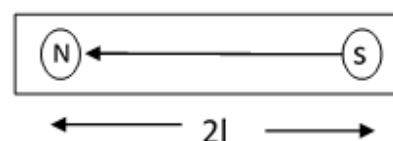
A current carrying straight solenoid behaves like a bar magnet. A study of magnetic field lines around a solenoid carrying current and magnetic field lines around a bar magnet shows that solenoid acts like a bar magnet.

- (i) Magnetic field inside a solenoid is stronger than inside a bar magnet.
- (ii) Magnetic field outside a solenoid is very weak.
- (iii) Magnetic field inside a solenoid is uniform, while outside it is non-uniform.



3. Magnetic dipole (bar magnet) in a uniform magnetic field :

A magnetic field dipole consists of a pair of magnetic poles equal and opposite strength separated by a small distance.



of

Dipole Moment of magnetic dipole-

The product of pole strength of either pole and distance between the magnetic pole.

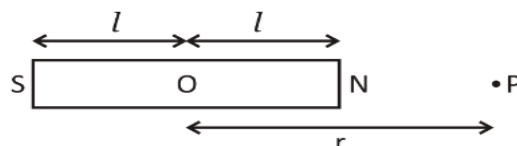
$$m = q_m \times 2l$$

S.I unit : Ampere-meter²

4. Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis

Magnetic field intensity due to a short bar magnet ($r \gg l$) along its axis

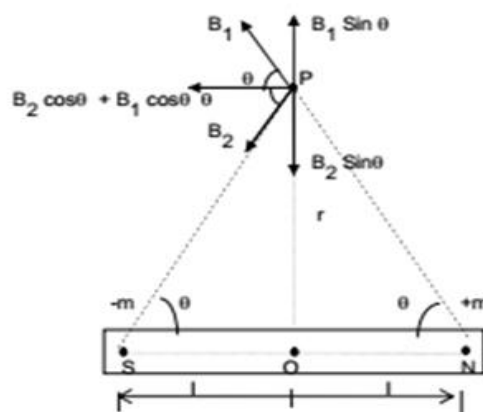
$$B_{axis} = \mu_0 \frac{2m}{4\pi r^3}$$



5. Magnetic field intensity due to a magnetic dipole (bar magnet) perpendicular to its axis

Magnetic field intensity due to a short bar magnet ($r \gg l$) perpendicular to its axis (at equatorial point)

$$B_{equatorial} = \mu_0 \frac{m}{4\pi r^3}$$



Note - $B_{axis} = 2 \times B_{equatorial}$

6. Torque on a magnetic dipole (bar magnet) in a uniform magnetic field

Torque acting on bar magnet

τ = magnitude of force $\times$ perpendicular dist.

$$\vec{\tau} = \vec{m} \times \vec{B} = mB \sin \theta \hat{n}$$

Special case -

(i) When $\theta = 0/\pi$, $\tau = 0$

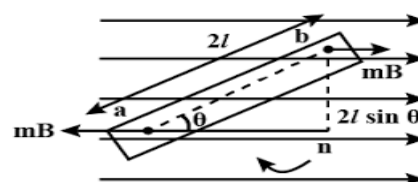
(ii) When $\theta = \frac{\pi}{2}$, $\tau = mB$ (maximum torque)

7. Potential energy of magnetic dipole in magnetic field

Work done to rotate a magnetic dipole in uniform magnetic field is stored as potential energy the magnetic dipole.

$$U = -\vec{M} \cdot \vec{B} = -MB \cos \theta$$

(i) When $\theta=0$, $U = -MB$ (minimum or stable equilibrium position)



- (ii) When $\theta = \pi$, $U = +MB$ (maximum or unstable equilibrium position)
 (iii) When $\theta = \frac{\pi}{2}$, potential energy is zero

8. Magnetic field lines

It is the space around a magnet within which its influence can be experienced by a small magnet.

(i) **Uniform Magnetic Field :**

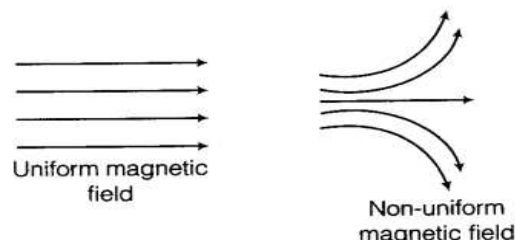
Magnetic field in a region is uniform if it has the same strength and same direction at all points in the region.

(ii) **Non uniform Magnetic Field :**

If it has different strength at different points in the region

Properties of Magnetic Field Lines:

- It is continuous and closed curves traveling from north pole to south pole outside but south pole to north pole inside the mag.
- The tangent at any point on the magnetic field line gives the direction of strength of the magnetic field.
- Two magnetic field lines do not intersect or cross each other because at the point of intersection there will be two directions at a single point, which is not possible.
- Widely spaced field lines represent weak magnetic field and closely spaced field lines represent strong magnetic field.
- Magnetic field lines are not real yet they represent a magnetic field which is real.



9. Magnetic flux

Number of magnetic field lines passing normally through a given surface. It is denoted by ϕ .

$$\phi = \vec{B} \cdot \vec{A}$$

SI unit- Weber.

Dimension = $[ML^2T^{-2}A^{-1}]$

10. Gauss's law in magnetism

It states that the surface integral of a magnetic field over a closed surface is always zero.

$$\oint \vec{B} \cdot d\vec{s} = 0$$

11. Some important terms

Magnetisation or Intensity of magnetisation :

The degree or extent to which a substance is magnetized when placed in the magnetizing field is called magnetisation denoted by I or M . The magnetic dipole moment per unit volume of the substance is known as intensity of magnetisation.

$$\vec{M} = \frac{\vec{m}}{V}$$

Intensity of Magnetizing Field :

The extent to which the magnetizing field can magnetize a substance is known as intensity of magnetizing field. Denoted by H .

$$H = \frac{B_0}{\mu_0}$$

Magnetic Permeability :

The extent to which magnetic field lines can enter a substance is known as magnetic permeability.

$$\mu = \frac{B}{H}$$

Relative Magnetic Permeability :

The ratio of the flux density inside the material to the flux density in vacuum (B_0) is known as relative magnetic permeability.

$$\mu_r = \frac{B}{B_0} = \frac{\mu}{\mu_0}$$

Magnetic Susceptibility :

It is the property of a substance which shows how easily the substance can be magnetized when placed in the magnetizing field denoted by χ_m

$$\chi_m = \frac{M}{H}$$

Relation between magnetic permeability and magnetic susceptibility :

$$\mu_r = 1 + \chi_m$$

12. Magnetic materials :

1. Diamagnetic Materials 2. Paramagnetic Materials 3. Ferro-magnetic Materials

1. Diamagnetic Materials :

These are the substances in which feeble magnetism is produced in a direction opposite to the applied magnetic field. These substances are repelled by a strong magnet. These substances have small negative values of susceptibility χ and positive low value of relative permeability μ_r .

$$-1 \leq \chi_m \leq 0 \quad \text{and} \quad 0 \leq \mu_r \leq 1$$

The examples of diamagnetic substances are bismuth, antimony, copper, lead, zinc, tin, gold, silicon, water, nitrogen (at STP) and sodium chloride.

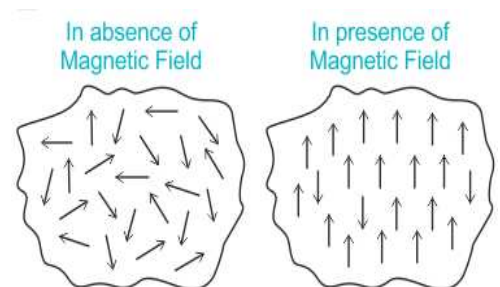
2. Paramagnetic Materials :

These are the substances in which feeble magnetism is induced in the same direction as the applied magnetic field. These are feebly attracted by a strong magnet. These substances have small positive values of M and χ and relative permeability μ_r greater than 1

$$0 \leq \chi_m \leq \varepsilon \quad \text{and} \quad 1 \leq \mu_r \leq 1 + \varepsilon$$

where ε is a small positive number. The examples of paramagnetic substances are platinum, chromium, sodium, aluminium, calcium, manganese, oxygen (at STP) and copper chloride.

Electron Theory :



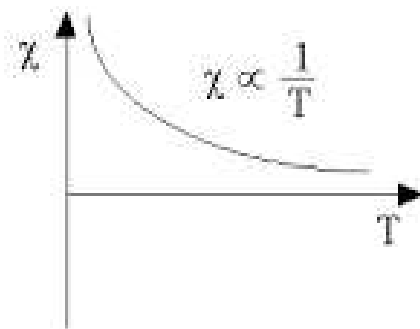
Paramagnetism

These materials have some permanent magnetic dipole moment. Due to thermal agitation they cancel the magnetic moments of each other. When a magnetic field is applied, atomic dipoles try to line up in the direction of and get weakly magnetized.

Curie's Law in Paramagnetism :

Curie's law states that magnetisation of a paramagnetic material is inversely proportional to absolute temperature (T) of the material.

$$\frac{M}{H} = \chi_m = \frac{\mu_0 C}{T} \Rightarrow \chi_m \propto \frac{1}{T}$$



3. Ferromagnetic Materials :

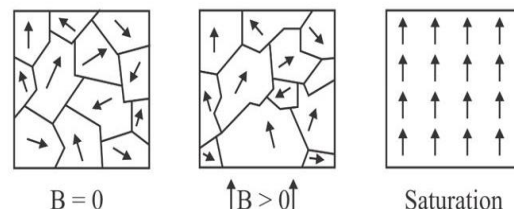
These are the substances in which a strong magnetism is produced in the same direction as the applied magnetic field. These are strongly attracted by a magnet. These substances are characterized by large positive values of M and χ and values of μ_r much greater than 1, eg. Iron, cobalt, nickel, gadolinium and alloy like alnico.

$$\chi_m \gg 1 \quad \text{and} \quad \mu_r \gg 1$$

Electron Theory :

Domain is a small region formed by a large number of atoms of a ferromagnetic substance. Electron of one atom interacts with the electron of a nearby atom known as coupling. So each domain possesses a strong magnetic moment. In the absence of an external magnetic field, these domains are randomly oriented so that the net magnetic moment of the substance is zero. When a magnetic field is applied, all domains align themselves along the direction of the field.

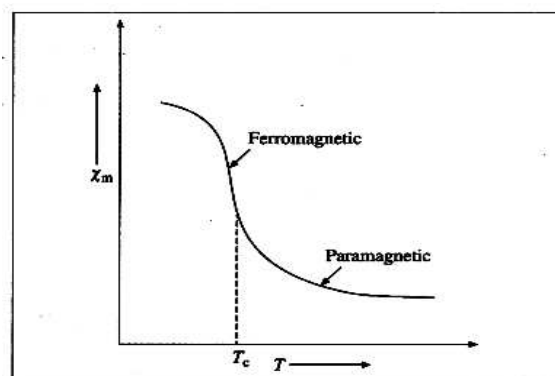
Ferromagnetism



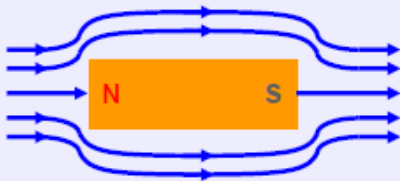
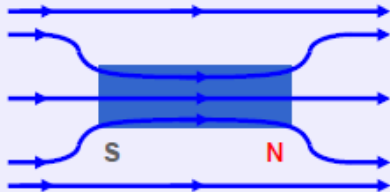
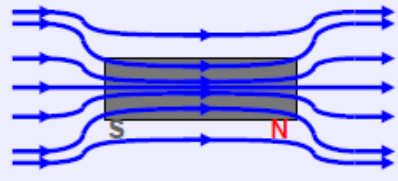
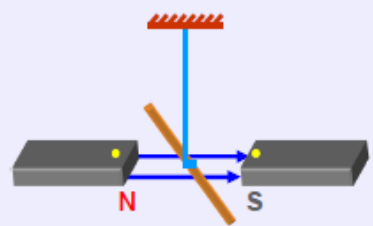
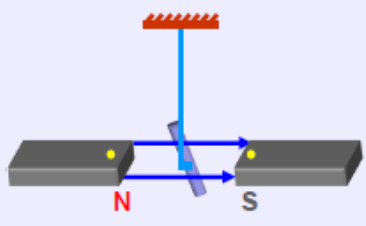
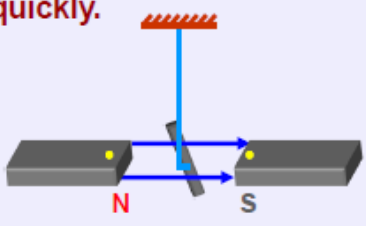
Curie's law of transition from ferromagnetism to paramagnetism :

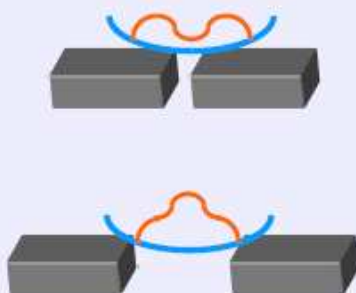
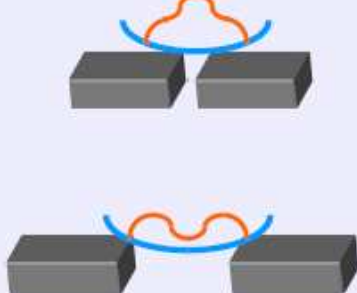
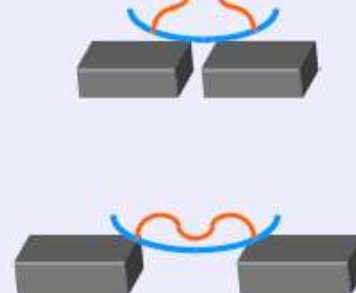
$$\chi_m = \frac{C}{T - T_c} \quad \text{where } T > T_c$$

When the temperature of ferromagnetic substance is greater than curie temperature then it is a paramagnetic substance.



Distinguish between Diamagnetism, Para magnetism and Ferromagnetism

DIA	PARA	FERRO
1. Diamagnetic substances are those substances which are feebly repelled by a magnet. Eg. Antimony, Bismuth, Copper, Gold, Silver, Quartz, Mercury, Alcohol, water, Hydrogen, Air, Argon, etc.	Paramagnetic substances are those substances which are feebly attracted by a magnet. Eg. Aluminium, Chromium, Alkali and Alkaline earth metals, Platinum, Oxygen, etc.	Ferromagnetic substances are those substances which are strongly attracted by a magnet. Eg. Iron, Cobalt, Nickel, Gadolinium, Dysprosium, etc.
2. When placed in magnetic field, the lines of force tend to avoid the substance. 	The lines of force prefer to pass through the substance rather than air. 	The lines of force tend to crowd into the specimen. 
3. When a diamagnetic rod is freely suspended in a uniform magnetic field, it aligns itself in a direction perpendicular to the field. 	When a paramagnetic rod is freely suspended in a uniform magnetic field, it aligns itself in a direction parallel to the field. 	When a paramagnetic rod is freely suspended in a uniform magnetic field, it aligns itself in a direction parallel to the field very quickly. 

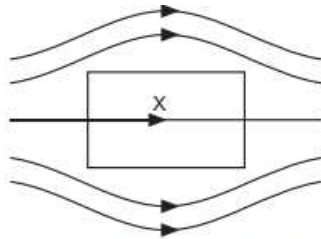
2. When placed in non-uniform magnetic field, it moves from stronger to weaker field (feeble repulsion).	When placed in non-uniform magnetic field, it moves from weaker to stronger field (feeble attraction).	When placed in non-uniform magnetic field, it moves from weaker to stronger field (strong attraction).
4. If diamagnetic liquid taken in a watch glass is placed in uniform magnetic field, it collects away from the centre when the magnetic poles are closer and collects at the centre when the magnetic poles are farther. 	If paramagnetic liquid taken in a watch glass is placed in uniform magnetic field, it collects at the centre when the magnetic poles are closer and collects away from the centre when the magnetic poles are farther. 	If ferromagnetic liquid taken in a watch glass is placed in uniform magnetic field, it collects at the centre when the magnetic poles are closer and collects away from the centre when the magnetic poles are farther. 
5. When a diamagnetic substance is placed in a magnetic field, it is weakly magnetised in the direction opposite to the inducing field.	When a paramagnetic substance is placed in a magnetic field, it is weakly magnetised in the direction of the inducing field.	When a ferromagnetic substance is placed in a magnetic field, it is strongly magnetised in the direction of the inducing field.
6. Induced Dipole Moment (M) is a small – ve value.	Induced Dipole Moment (M) is a small + ve value.	Induced Dipole Moment (M) is a large + ve value.

7. Intensity of Magnetisation (I) has a small – ve value.	Intensity of Magnetisation (I) has a small + ve value.	Intensity of Magnetisation (I) has a large + ve value.
8. Magnetic permeability μ is always less than unity.	Magnetic permeability μ is more than unity.	Magnetic permeability μ is large i.e. much more than unity.
9. Magnetic susceptibility c_m has a small – ve value.	Magnetic susceptibility c_m has a small + ve value.	Magnetic susceptibility c_m has a large + ve value.
10. They do not obey Curie's Law. i.e. their properties do not change with temperature.	They obey Curie's Law. They lose their magnetic properties with rise in temperature.	They obey Curie's Law. At a certain temperature called Curie Point, they lose ferromagnetic properties and behave like paramagnetic substances.

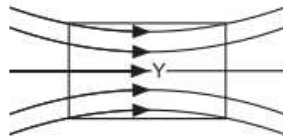
S. No.	Property	Diamagnetic	Paramagnetic	Ferromagnetic	Remark
1	Magnetic induction B	$B < B_0$	$B > B_0$	$B \gg B_0$	B_0 is magnetic induction in free space
2	Intensity of magnetization M	Small and negative	Small and positive	Very high and positive	
3	Magnetic susceptibility χ	Small and negative	Small and positive	Very high and positive	
4	Relative permeability μ_r	$\mu_r < 1$	$\mu_r > 1$	$\mu_r \gg 1$ (of the order of thousand)	

GRAPHS :-

- Magnetic field lines when diamagnetic and paramagnetic substances are placed in external magnetic field



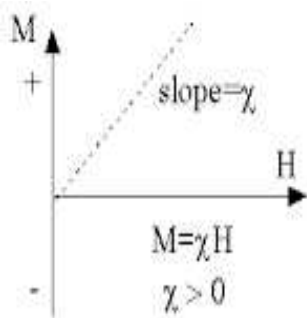
(a) Diamagnetic substance



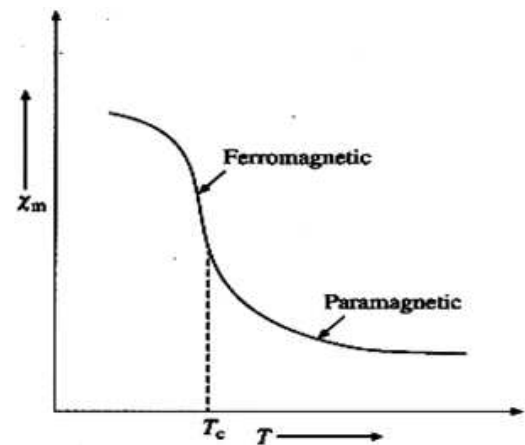
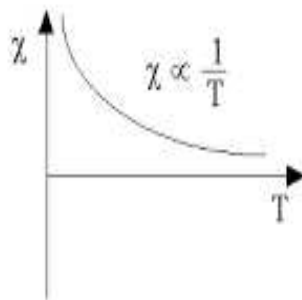
(b) Paramagnetic substance

- Effect of temperature on Paramagnetic material (Curie's law)

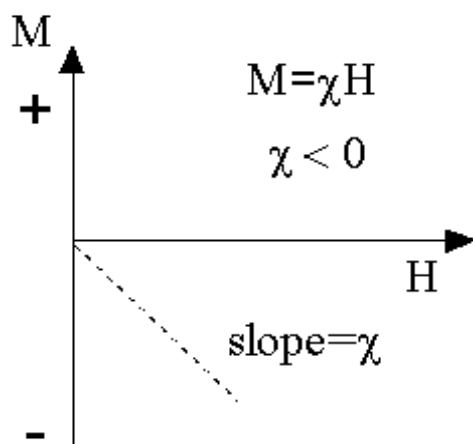
Ferromagnetic material



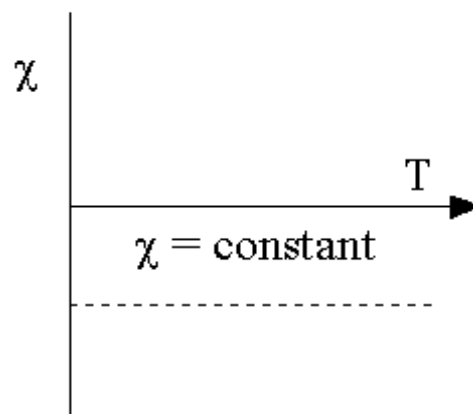
Paramagnetism



- Effect of temperature on diamagnetic material



Diamagnetism



UNITS :-

Physical quantity	Symbol	Nature	Dimensions	Units	Remarks
Permeability of free space	μ_0	Scalar	$[MLT^{-2}A^{-2}]$	Henry/meter Or TmA^{-1}	$4\pi \times 10^{-7}$
Magnetic field	B	Vector	$[MT^{-2}A^{-1}]$	T (tesla) or N/A-m or Wb/m^2	
Magnetic moment	m	Vector	$[L^2A]$	$A\cdot m^2$ or J/T	
Magnetic flux	ϕ	Scalar	$[ML^2T^{-2}A^{-1}]$	Weber	
Magnetisation	M	Vector	$[L^{-1}A]$	Am^{-1}	$\frac{\text{Magnetic moment}}{\text{volume}}$
Magnetic intensity	H	Vector	$[L^{-1}A]$	Am^{-1}	
Magnetic susceptibility	χ	Scalar	Dimensionless	unitless	$\chi = \frac{M}{H}$
Relative magnetic permeability	μ_r	Scalar	Dimensionless	unitless	$B = \mu_0\mu_r H$
magnetic permeability	μ	Scalar	$[MLT^{-2}A^{-2}]$	NA^{-2} Or TmA^{-1}	$\mu = \mu_0\mu_r$